VT100 Escape Codes

This document describes how to control a VT100 terminal. Entries are of the form "name, description, escape code".

Escape codes begin with ESC (represented as `^[`).

Source: `https://espterm.github.io/docs/VT100%20escape%20codes.html`

VT100 / ANSI Mode Escape Codes

Name	Description	`Esc Code`
setnl LMN	Set new line mode	`^[[20h`
setappl DECCKM	Set cursor key to application	`^[[?1h`
setansi DECANM	Set ANSI (versus VT52)	`none`
setcol DECCOLM	Set number of columns to 132	`^[[?3h`
setsmooth DECSCLM	Set smooth scrolling	`^[[?4h`
setrevscrn DECSCNM	Set reverse video on screen	`^[[?5h`
setorgrel DECOM	Set origin to relative	`^[[?6h`
setwrap DECAWM	Set auto-wrap mode	`^[[?7h`
setrep DECARM	Set auto-repeat mode	`^[[?8h`
setinter DECINLM	Set interlacing mode	`^[[?9h`
setlf LMN	Set line feed mode	`^[[20l`
setcursor DECCKM	Set cursor key to cursor	`^[[?1l`
setvt52 DECANM	Set VT52 (versus ANSI)	`^[[?2l`
resetcol DECCOLM	Set number of columns to 80	`^[[?3l`
setjump DECSCLM	Set jump scrolling	`^[[?4l`
setnormscrn DECSCNM	Set normal video on screen	`^[[?5l`

Name	Description	Esc Code
setorgabs DECOM	Set origin to absolute	^[[?6l
resetwrap DECAWM	Reset auto-wrap mode	^[[?7l
resetrep DECARM	Reset auto-repeat mode	^[[?8l
resetinter DECINLM	Reset interlacing mode	^[[?9l
altkeypad DECKPAM	Set alternate keypad mode	^[=
numkeypad DECKPNM	Set numeric keypad mode	^[>
setukg0	Set United Kingdom G0 character set	^[(A
setukg1	Set United Kingdom G1 character set	^[)A
setusg0	Set United States G0 character set	^[(B
setusg1	Set United States G1 character set	^[)B
setspecg0	Set G0 special chars. & line set	^[(0
setspecg1	Set G1 special chars. & line set	^[)0
setaltg0	Set G0 alternate character ROM	^[(1
setaltg1	Set G1 alternate character ROM	^[)1
setaltspecg0	Set G0 alt char ROM and spec. graphics	^[(2
setaltspecg1	Set G1 alt char ROM and spec. graphics	^[)2
setss2 SS2	Set single shift 2	^[N
setss3 SS3	Set single shift 3	^[O
modesoff SGR0	Turn off character attributes	^[[m
modesoff SGR0	Turn off character attributes	^[[0m
bold SGR1	Turn bold mode on	^[[1m
lowint SGR2	Turn low intensity mode on	^[[2m
underline SGR4	Turn underline mode on	^[[4m
blink SGR5	Turn blinking mode on	^[[5m

Name	Description	Esc Code
reverse SGR7	Turn reverse video on	^[[7m
invisible SGR8	Turn invisible text mode on	^[[8m
setwin DECSTBM	Set top and bottom line#s of a window	^[[<v>;<v>r
cursorup(n) CUU	Move cursor up n lines	^[[<n>A
cursordn(n) CUD	Move cursor down n lines	^[[<n>B
cursorrt(n) CUF	Move cursor right n lines	^[[<n>C
cursorlf(n) CUB	Move cursor left n lines	^[[<n>D
cursorhome	Move cursor to upper left corner	^[[H
cursorhome	Move cursor to upper left corner	^[[;H
cursorpos(v,h) CUP	Move cursor to screen location v,h	^[[<v>;<h>H
hvhome	Move cursor to upper left corner	^[[f
hvhome	Move cursor to upper left corner	^[[;f
hvpos(v,h) CUP	Move cursor to screen location v,h	^[[<v>;<h>f
index IND	Move/scroll window up one line	^[D
revindex RI	Move/scroll window down one line	^[M
nextline NEL	Move to next line	^[E
savecursor DECSC	Save cursor position and attributes	^[7
restorecursor DECSC	Restore cursor position and attributes	^[8
tabset HTS	Set a tab at the current column	^[H
tabclr TBC	Clear a tab at the current column	^[[g
tabclr TBC	Clear a tab at the current column	^[[0g
tabclrall TBC	Clear all tabs	^[[3g
dhtop DECDHL	Double-height letters, top half	^[#3
dhbot DECDHL	Double-height letters, bottom half	^[#4

Name	Description	Esc Code
swsh DECSWL	Single width, single height letters	^[#5
dwsh DECDWL	Double width, single height letters	^[#6
cleareol EL0	Clear line from cursor right	^[[K
cleareol EL0	Clear line from cursor right	^[[0K
clearbol EL1	Clear line from cursor left	^[[1K
clearline EL2	Clear entire line	^[[2K
cleareos ED0	Clear screen from cursor down	^[[J
cleareos ED0	Clear screen from cursor down	^[[0J
clearbos ED1	Clear screen from cursor up	^[[1J
clearscreen ED2	Clear entire screen	^[[2J
devstat DSR	Device status report	^[5n
termok DSR	Response: terminal is OK	^[0n
termnok DSR	Response: terminal is not OK	^[3n
getcursor DSR	Get cursor position	^[6n
cursorpos CPR	Response: cursor is at v,h	^[<v>;<h>R
ident DA	Identify what terminal type	^[[c
ident DA	Identify what terminal type (another)	^[[0c
gettype DA	Response: terminal type code n	^[[?1;<n>0c
reset RIS	Reset terminal to initial state	^[c
align DECALN	Screen alignment display	^[#8
testpu DECTST	Confidence power up test	^[[2;1y
testlb DECTST	Confidence loopback test	^[[2;2y
testpurep DECTST	Repeat power up test	^[[2;9y
testlbrep DECTST	Repeat loopback test	^[[2;10y

Name	Description	Esc Code
ledsoff DECLL0	Turn off all four leds	^[[0q
led1 DECLL1	Turn on LED #1	^[[1q
led2 DECLL2	Turn on LED #2	^[[2q
led3 DECLL3	Turn on LED #3	^[[3q
led4 DECLL4	Turn on LED #4	^[[4q

VT52 Compatibility Mode Codes

All codes below are for use in VT52 compatibility mode.

Name	Description	Esc Code
setansi	Enter/exit ANSI mode (VT52)	^[<
altkeypad	Enter alternate keypad mode	^[=
numkeypad	Exit alternate keypad mode	^[>
setgr	Use special graphics character set	^[F
resetgr	Use normal US/UK character set	^[G
cursorup	Move cursor up one line	^[A
cursordn	Move cursor down one line	^[B
cursorrt	Move cursor right one char	^[C
cursorlf	Move cursor left one char	^[D
cursorhome	Move cursor to upper left corner	^[H
cursorpos(v,h)	Move cursor to v,h location	^[<v><h>
revindex	Generate a reverse line-feed	^[I
cleareol	Erase to end of current line	^[K
cleareos	Erase to end of screen	^[J
ident	Identify what the terminal is	^[Z
identresp	Correct response to ident	^[/Z

VT100 Special Key Codes

These are sent from the terminal back to the computer when the particular key is pressed.

Numeric keypad keys send different codes in numeric mode than in alternate mode.

Function Keys

Key	`Esc Code`
PF1	`^[OP`
PF2	`^[OQ`
PF3	`^[OR`
PF4	`^[OS`

Arrow Keys

Key	Reset	`Set`
Up	^[A	`^[OA`
Down	^[B	`^[OB`
Right	^[C	`^[OC`
Left	^[D	`^[OD`

Numeric Keypad Keys

Key	Numeric	`Alternate`
0	0	`^[Op`
1	1	`^[Oq`

Key	Numeric	Alternate
2	2	^[Or
3	3	^[Os
4	4	^[Ot
5	5	^[Ou
6	6	^[Ov
7	7	^[Ow
8	8	^[Ox
9	9	^[Oy
- (minus)	-	^[Om
, (comma)	,	^[Ol
. (period)	.	^[On
ENTER	^M	^[OM

ESC character values

ESC: `27` (decimal), `033` (octal), `0x1b` (hexadecimal).